

CE 310 — Week 6 In-Class Exercise

Empirical Percentiles Across CE Practice

Total: 100 points (Section A: 16 · Section B: 60 · Chart 1: 7 · Chart 2: 8 · Memo: 9)

Before You Begin

Dataset 1: Week5_CE_Measurements.csv — the same file from Week 5.

Columns (Dataset 1): A=RecordID · B=Supplier (“Alpha”/“Beta”) · C=TimeOfDay (“AM_Peak”/“PM_Peak”) · D=fc_psi · E=Speed_mph · F=CostPerCY

Dataset 2 (new): Week6_Noise_Measurements.csv — A=RecordID · B=Location (“Road-side”/“Courtyard”) · C=NoiseLevel_dB

Answer sheet: the Week 6 in-class answer sheet — sheets Answers and Written. Do not rename, reorder, or delete any sheet. The assignment has its own file, the Week 6 assignment answer sheet.

Cell layout on the Answers sheet: Column A = question # (locked) · Column B = question text (locked) · Column C = your formula/approach · Column D = intermediate value (★ questions only) · Column E = your numerical answer · Column F = unit (locked).

Type your formula as text. In Col C, put an apostrophe before the = so Excel stores the formula instead of running it — for example, '=AVERAGE(B2:B8761). Without the apostrophe Excel evaluates it and your method is lost. Screenshots are not accepted; the typed formula is what earns the approach credit.

Background — From Week 5 to Week 6

Last week you fit a Normal model (concrete, speed) or lognormal model (cost) to this same dataset and used NORM.INV/LOGNORM.DIST to estimate percentiles. Today you compute the actual empirical percentiles — no model, no assumption about shape — and compare.

Setup (≈10 min): Open both datasets, confirm =COUNTA(A:A)-1 returns 500 for each, confirm your column letters match the tables above.

Section A — Concrete Strength and Site Acoustics (16 pts)

Same four-question toolkit — LARGE, SMALL, MEDIAN, COUNTIF — applied to two independent domains. Round to the nearest 0.01 unless the question says otherwise.

Concrete (Dataset 1, column D fc_psi, all 500 records):

A1 (2 pts). The 90th percentile strength, using the exact-rank method. Hint: LARGE, $k = \text{ROUND}(500 \times 0.10, 0)$.

A2 (2 pts). The 10th percentile strength, using the exact-rank method — this is the value ACI 318 cares about. Hint: SMALL, same k as A1.

A3 (2 pts). What percentage of all 500 tests fall below 4,000 psi? Hint: COUNTIF, remember to divide by 500 and multiply by 100.

A4 (2 pts). The empirical median strength.

Site acoustics (Dataset 2, column C NoiseLevel_dB, all 500 records):

A5 (2 pts). The L10 (level exceeded 10% of the time — “peak” noise), using the exact-rank method. Hint: LARGE, same k logic as A1 — L10 is the 90th percentile of the readings.

A6 (2 pts). The L90 (level exceeded 90% of the time — “background” noise), using the exact-rank method. Hint: SMALL, same k logic as A2 — L90 is the 10th percentile of the readings.

A7 (2 pts). The empirical median noise level (L50).

A8 (2 pts). What percentage of all 500 readings exceed 65 dB — the HUD site-acceptability threshold? Hint: COUNTIF, same pattern as A3.

Section B — Traffic Speed and Construction Cost (60 pts)

Traffic Speed (column E) — all 500 records unless a subgroup is specified

B1 (4 pts). The 85th-percentile speed for all records, using the exact-rank method — this is the value the 85th-percentile speed rule (Part 1 of today’s lecture) is built on. Hint: LARGE, $k = \text{ROUND}(500 \times 0.15, 0)$.

B2★ (5 pts). The 10th-percentile speed for all records, using the exact-rank method. Column D: what k did you use? Hint: SMALL.

B3 (4 pts). Week 5’s Normal model estimated the 85th-percentile speed at 62.5 mph. What is the gap between your B1 answer and 62.5 ($B1 - 62.5$)?

B4 (4 pts). What percentage of all records show speed above 65 mph?

B5 (4 pts). The 25th percentile (Q1) speed for all records.

B6 (4 pts). The 75th percentile (Q3) speed for all records.

Target shape for Chart 1

Figure 1: Target shape for Chart 1

IQR diagnostic (ungraded): Compute $IQR = B6 - B5$. By the Normal approximation, $IQR \div 1.35 \approx \sigma$. Compare this robust estimate to your Week 5 standard deviation (~ 7.8 mph for all records). Close agreement means the distribution is approximately Normal through the middle; a larger gap suggests the tails depart from the Normal model.

B7★ (5 pts). The 85th-percentile speed, AM Peak only. Column D: how many AM Peak records are there? (Use COUNTIF on column C.) Hint: LARGE(FILTER(...)) or the array-formula equivalent.

B8 (4 pts). The 85th-percentile speed, PM Peak only, using the interpolated method. Hint: PERCENTILE(FILTER(...)).

B9 (4 pts). What percentage of all records show speed above 45 mph? (Week 5's Normal model estimated 88.5% for this same question — you are not required to enter the comparison, just the empirical value.)

Construction Cost (column F) — the AACE contingency question

B10★ (5 pts). The 90th-percentile cost, using the exact-rank method — this is the value AACE-style contingency budgeting is built on. Column D: what k did you use? Hint: LARGE, $k = \text{ROUND}(500 \times 0.10, 0)$.

B11 (4 pts). The 90th-percentile cost, using the interpolated method. Compare to B10. Hint: PERCENTILE.

B12 (4 pts). Week 5's lognormal model estimated the 90th-percentile cost at \$289.61/CY. What is the gap between your B10 answer and 289.61 ($B10 - 289.61$)?

B13★ (5 pts). The total dollar exposure across all records whose cost exceeds \$250/CY — i.e., $\sum \max(\text{Cost} - 250, 0)$. Column D: how many records exceed \$250? Hint: SUMPRODUCT.

B14 (4 pts). The median cost for Supplier Beta only. Hint: MEDIAN(FILTER(...)).

Required Charts

Chart 1 (7 pts) — Concrete Strength Duration Curve. Build Rank (column H, 1–500) and a sorted fc_psi column (column I, using LARGE(\$D\$2:\$D\$501, H2) filled down). Plot as a line chart, Rank on x-axis, strength on y-axis. Title: “Concrete Compressive Strength Duration Curve (n = 500)”. Add a horizontal reference line at your A2 answer (the ACI 318 threshold).

Target shape for Chart 2

Figure 2: Target shape for Chart 2

Chart 2 (8 pts) — Empirical vs. Normal Model, Speed. Build a Percentile column (0.01 to 0.99, 99 rows). Empirical column: =PERCENTILE(\$E\$2:\$E\$501, A2). Normal-model column: =NORM.INV(A2, 54.41, 7.81). Scatter chart with smooth lines, both series, legend. Title: “Vehicle Speed: Empirical vs. Normal Model”.

Memo (9 pts, in the Written sheet)

Write 3 short paragraphs:

1. Paragraph 1 (3 pts). Report your A2 (empirical 10th-pct concrete strength) and compare it to Week 5’s Normal-model estimates for Supplier Alpha (4,191 psi) and Supplier Beta (3,747 psi) individually — cite the actual Alpha/Beta empirical values from lecture (4,181 / 3,688 psi) and state which supplier(s) would fail a 4,000 psi spec.
 2. Paragraph 2 (3 pts). Report your B13 (total cost exposure above \$250/CY) and explain what it tells a project manager that the B10/B11 percentile value alone does not.
 3. Paragraph 3 (3 pts). Compare your B1 (all-records 85th-pct speed) to B7 and B8 (AM/PM conditional). Cite at least 2 numbers. What would you recommend a traffic engineer do differently for AM vs. PM if this were a real corridor?
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Before You Submit

- ☐ Full name and NetID entered on the Answer Sheet
- ☐ File saved as .xlsx and named CE310_W6_<NetID>.xlsx
- ☐ Sheet tabs not renamed, reordered, or deleted
- ☐ Every answer typed into its designated cell in Col E — answers placed anywhere else receive no credit
- ☐ Your formula typed as text in Col C, with an apostrophe before the = — screenshots are not accepted
- ☐ ★ questions also have their intermediate value in Col D
- ☐ Both charts built, with titles and axis labels
- ☐ Memo written in the Written sheet
- ☐ Uploaded to D2L before the deadline

Grading Summary: Section A (16) + Section B (60) = 76. Chart 1 (7) + Chart 2 (8) + Memo (9) = 24. Total: 100 pts.

Pacing: Setup + Section A + Section B1–B9 are the in-class target (~60 min). B10–B14, both charts, and the memo may extend into take-home time — check with your preceptor.

Excel Quick Reference

Concept	Formula
kth largest / smallest	=LARGE(range, k) / =SMALL(range, k)
Median	=MEDIAN(range)
Interpolated percentile	=PERCENTILE(range, p)
Exceedance probability	=COUNTIF(range, ">x") / 500 * 100
Band probability (between two values)	=COUNTIFS(range, ">lower", range, "<upper") / 500 * 100
Conditional percentile	=PERCENTILE(FILTER(range, condition), p)
Total exposure above threshold	=SUMPRODUCT((range-threshold)*(range>threshold))